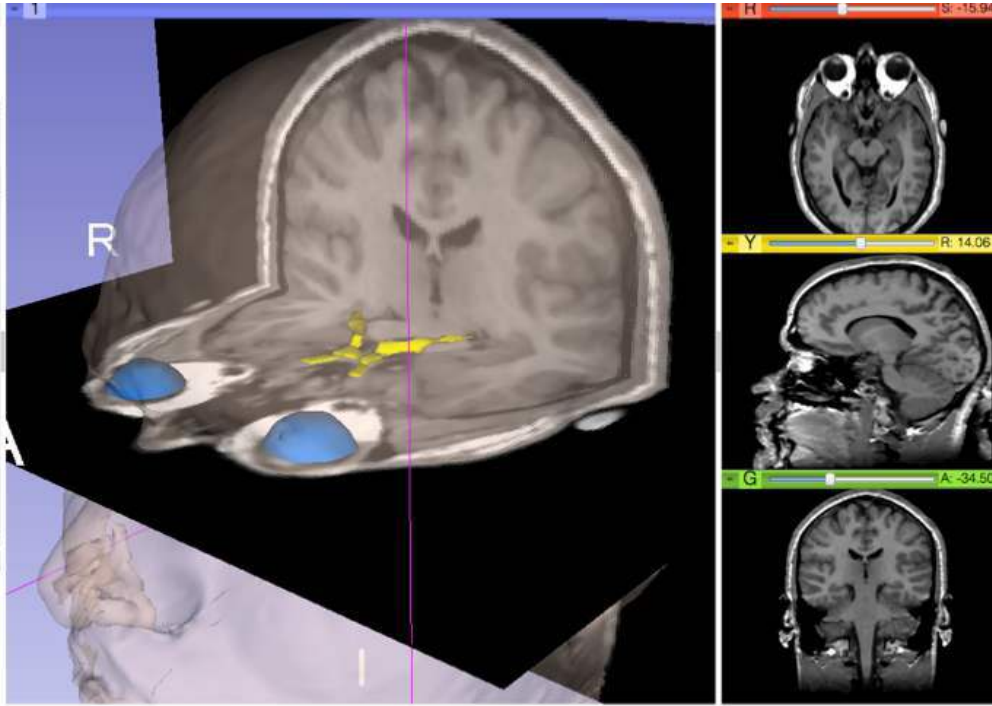


Slicer4 Minute

Sonia Pujol, Ph.D.

Assistant Professor of Radiology
Brigham and Women's Hospital
Harvard Medical School

Slicer4 minute tutorial

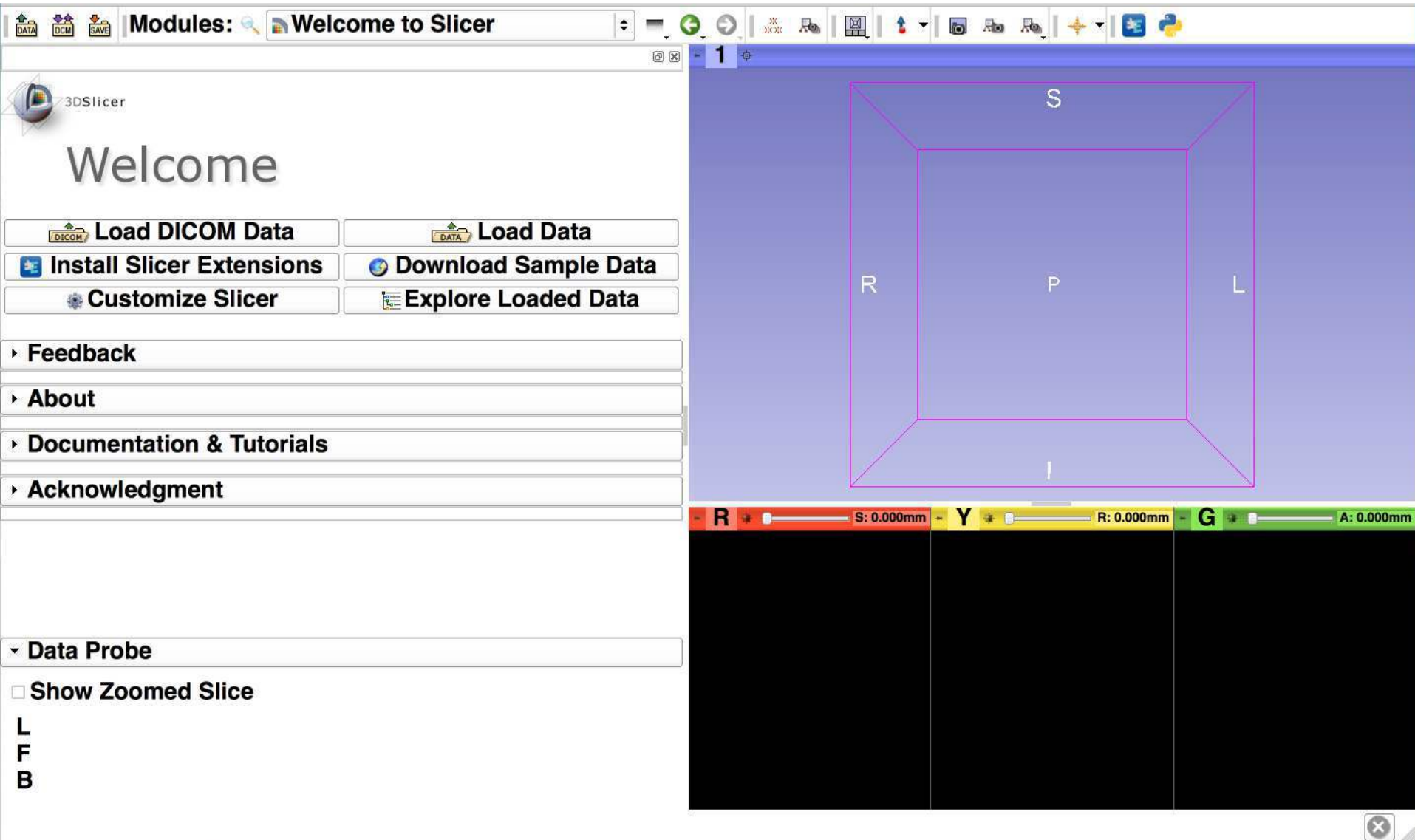


This tutorial is a 4-minute introduction to the 3D visualization capabilities of the Slicer3 software for medical image analysis.

Slicer4 software & dataset

- Download the Slicer4 software available at <http://download.slicer.org/>
- Download the Slicer4minute dataset available at <https://www.slicer.org/wiki/Documentation/4.10/Training>

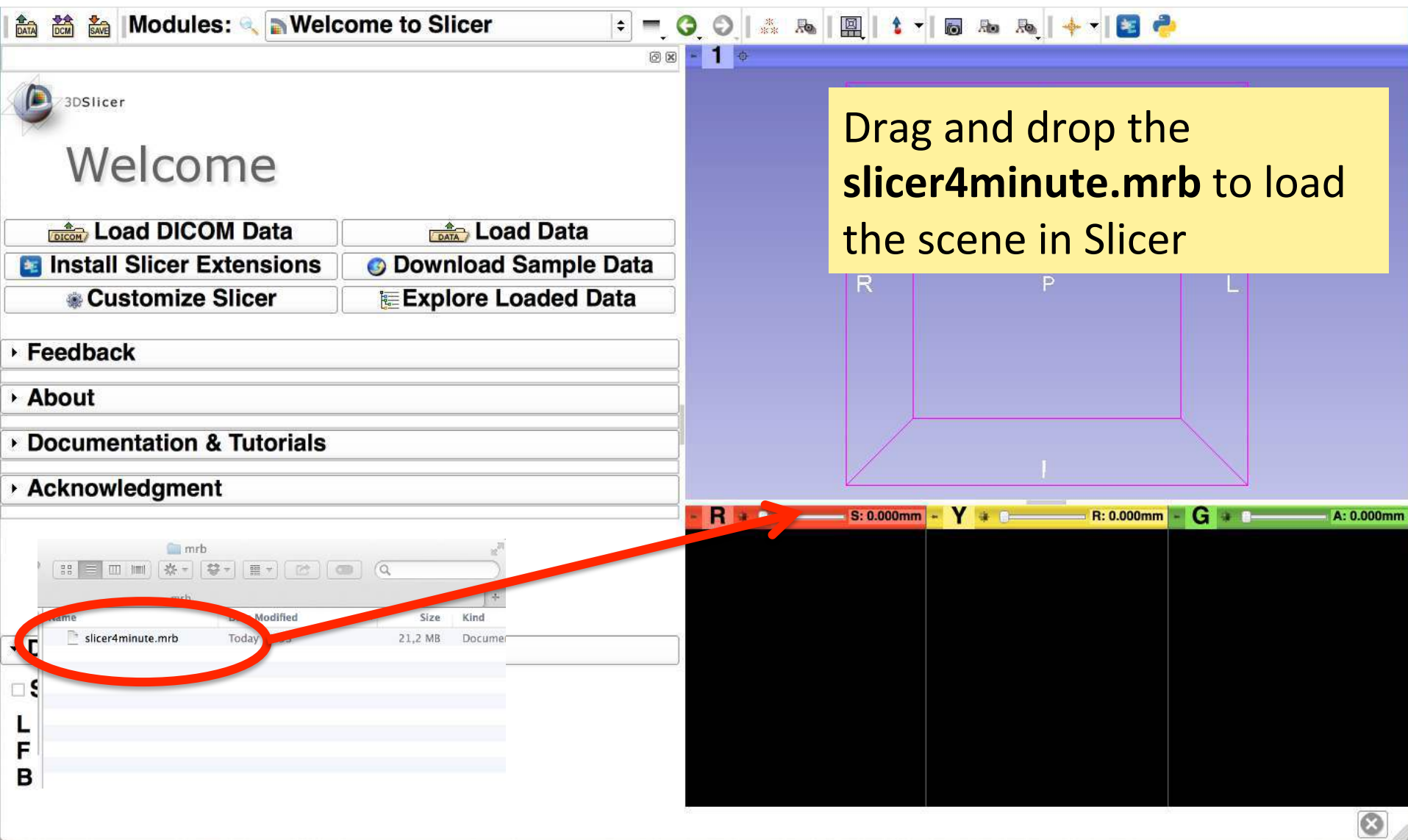
3D Slicer version 4



3D Slicer Scene

- A Slicer scene is a **MRML (Medical Reality Modeling Language)** file that contains a list of elements loaded into Slicer (volumes, models, fiducials, transforms, etc.)
- In the following example, we use a scene 'Slicer4minute.mrml' composed of an MRI scan and 3D models of the head.
- The scene file and datasets have been saved as a **MRB** (Medical Reality Bundle) file.
- The MRB file format is Slicer's archive file format.

Loading the Slicer4minute dataset



The image shows the 3D Slicer software interface. The main window displays the 'Welcome' screen with various options like 'Load DICOM Data', 'Load Data', 'Install Slicer Extensions', 'Download Sample Data', 'Customize Slicer', and 'Explore Loaded Data'. A file explorer window is open, showing a folder named 'mrb' containing a file named 'slicer4minute.mrb'. A red circle highlights this file, and a red arrow points from it to the 'R' button in the top toolbar. A yellow text box in the center of the main window provides instructions on how to load the dataset.

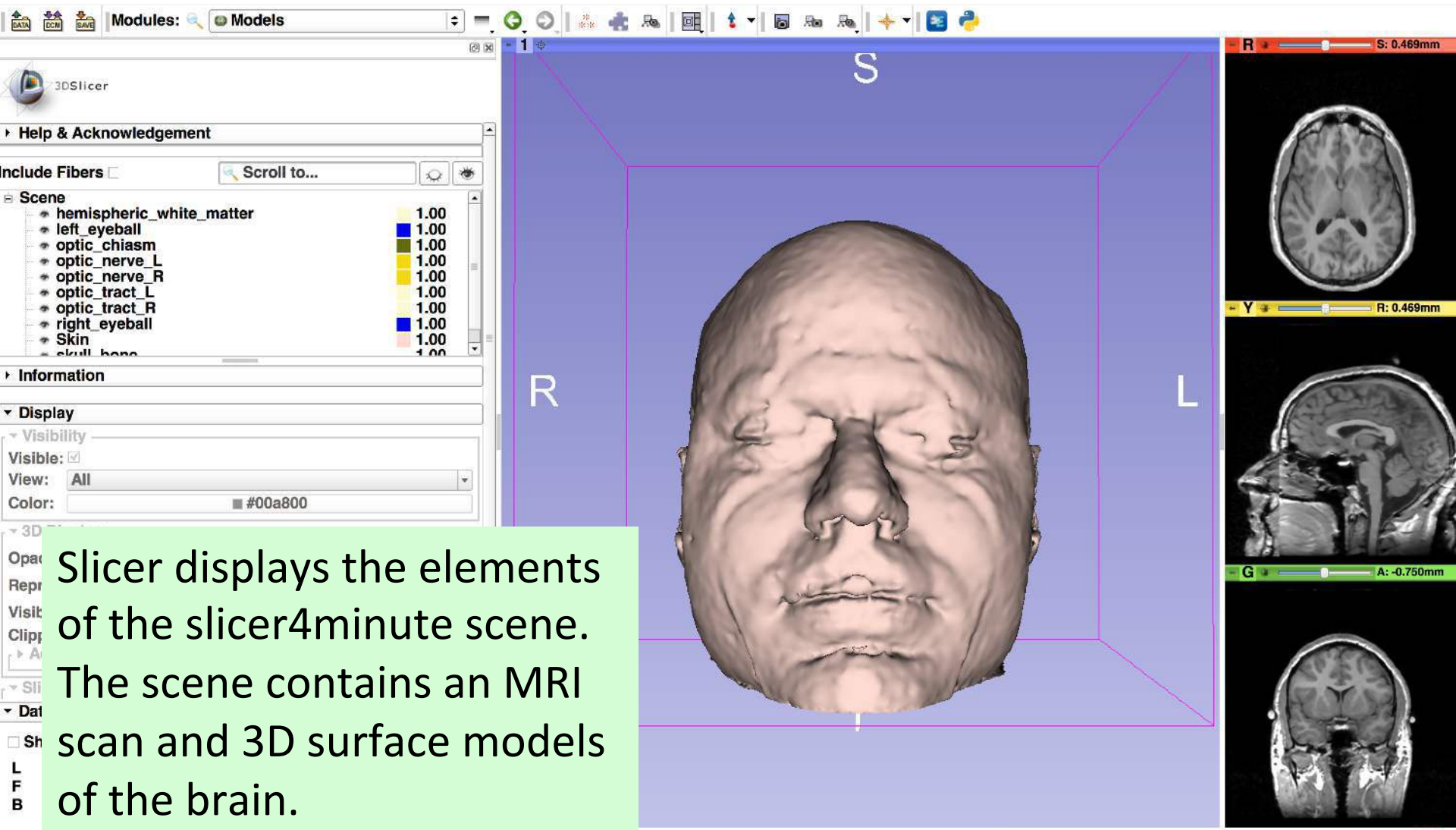
Drag and drop the **slicer4minute.mrb** to load the scene in Slicer

File Explorer (mrb):

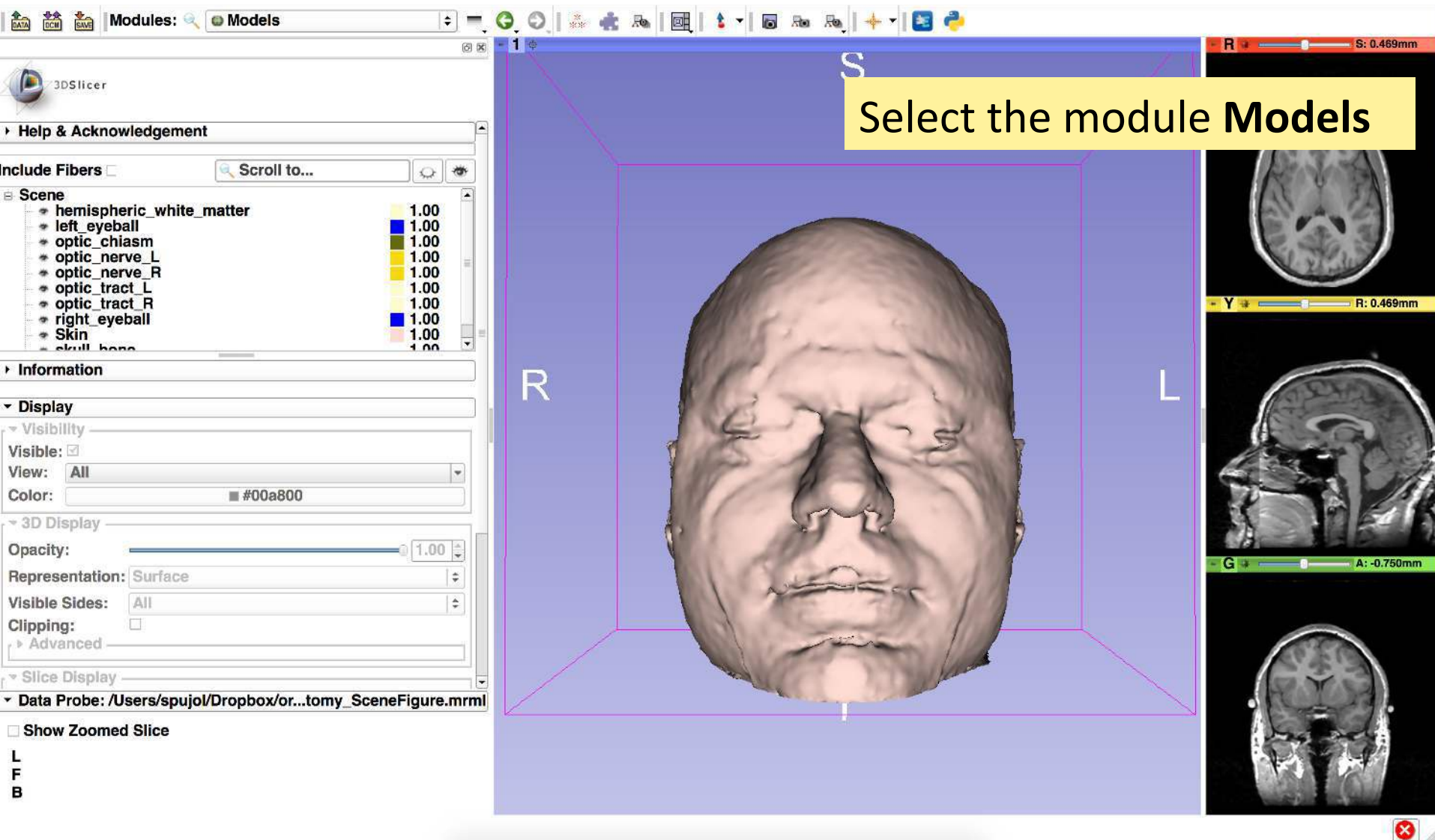
Name	Modified	Size	Kind
slicer4minute.mrb	Today	21,2 MB	Document

Top Toolbar: R S: 0.000mm Y R: 0.000mm G A: 0.000mm

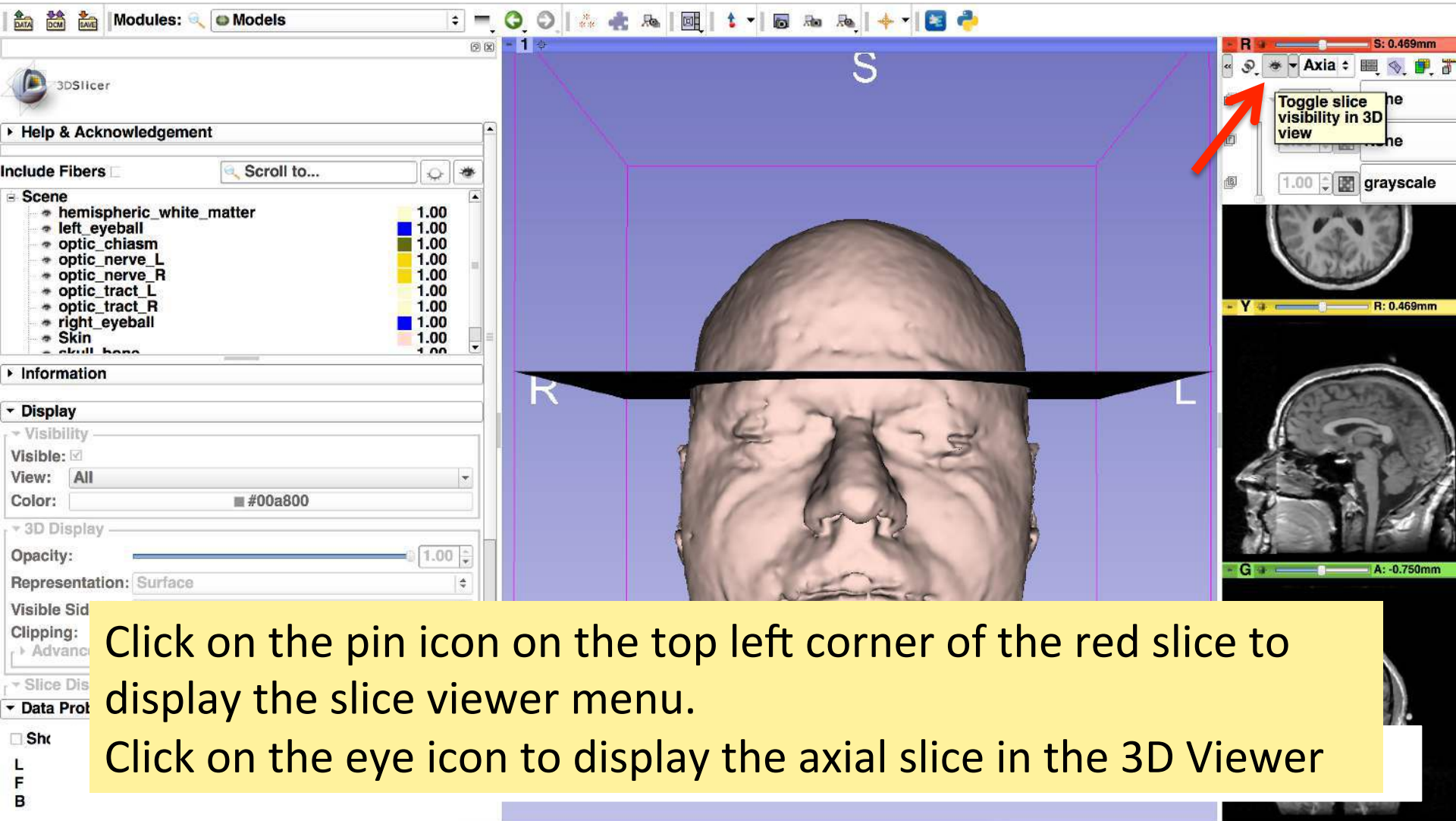
Slicer4minute Scene



3D Visualization



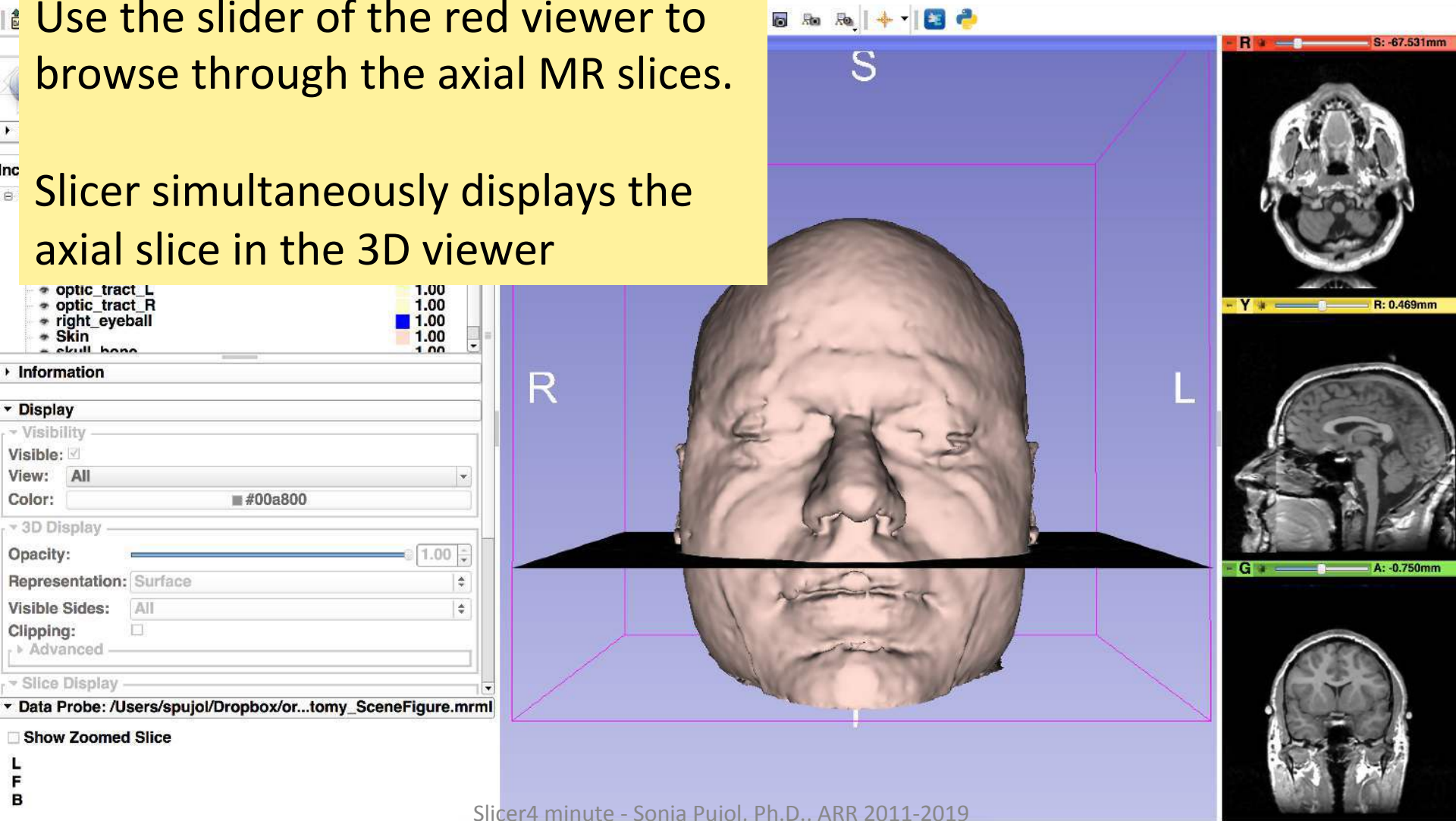
3D Visualization



3D Visualization

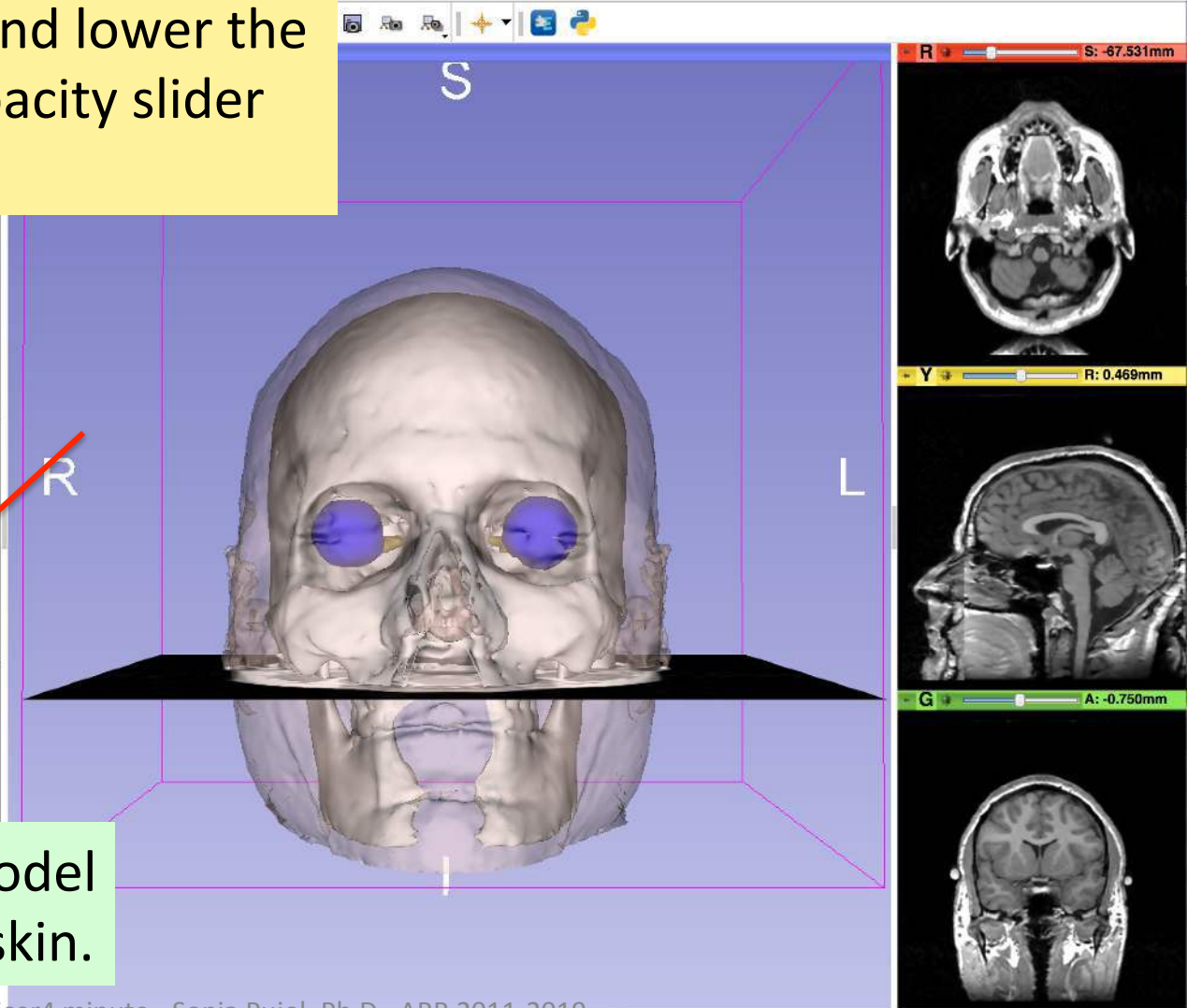
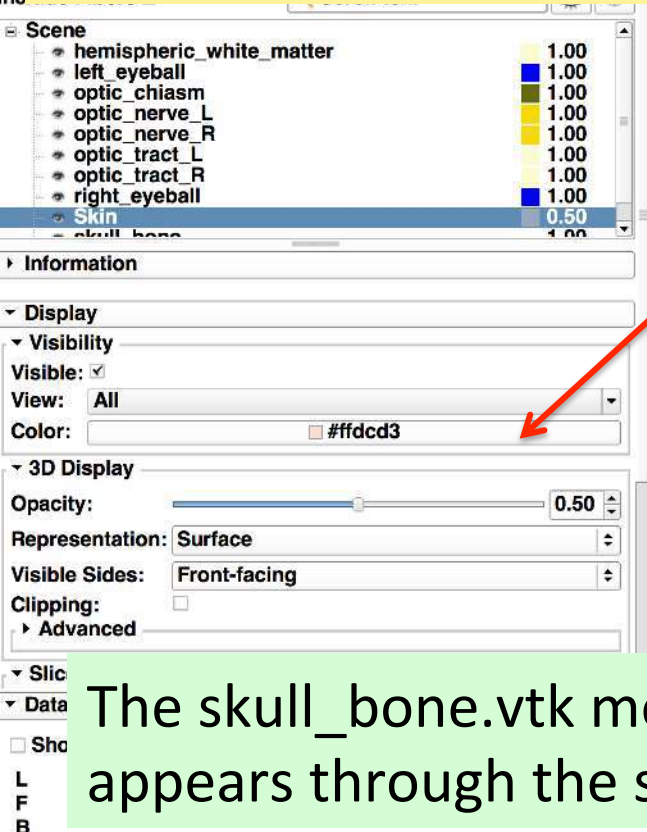
Use the slider of the red viewer to browse through the axial MR slices.

Slicer simultaneously displays the axial slice in the 3D viewer



3D Visualization

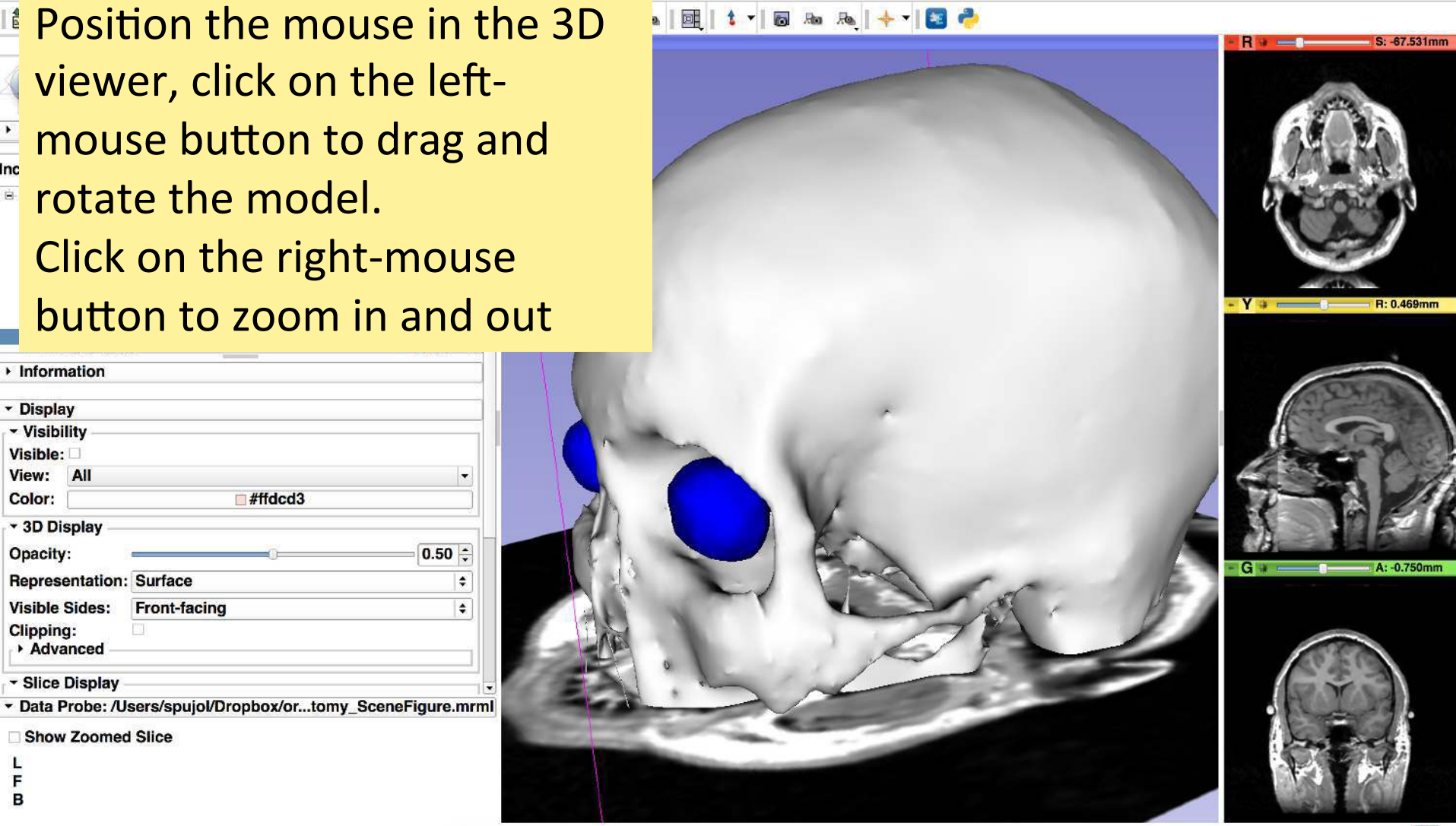
Select the Skin model and lower the its opacity using the Opacity slider in the 3D Display tab



The skull_bone.vtk model appears through the skin.

3D Visualization

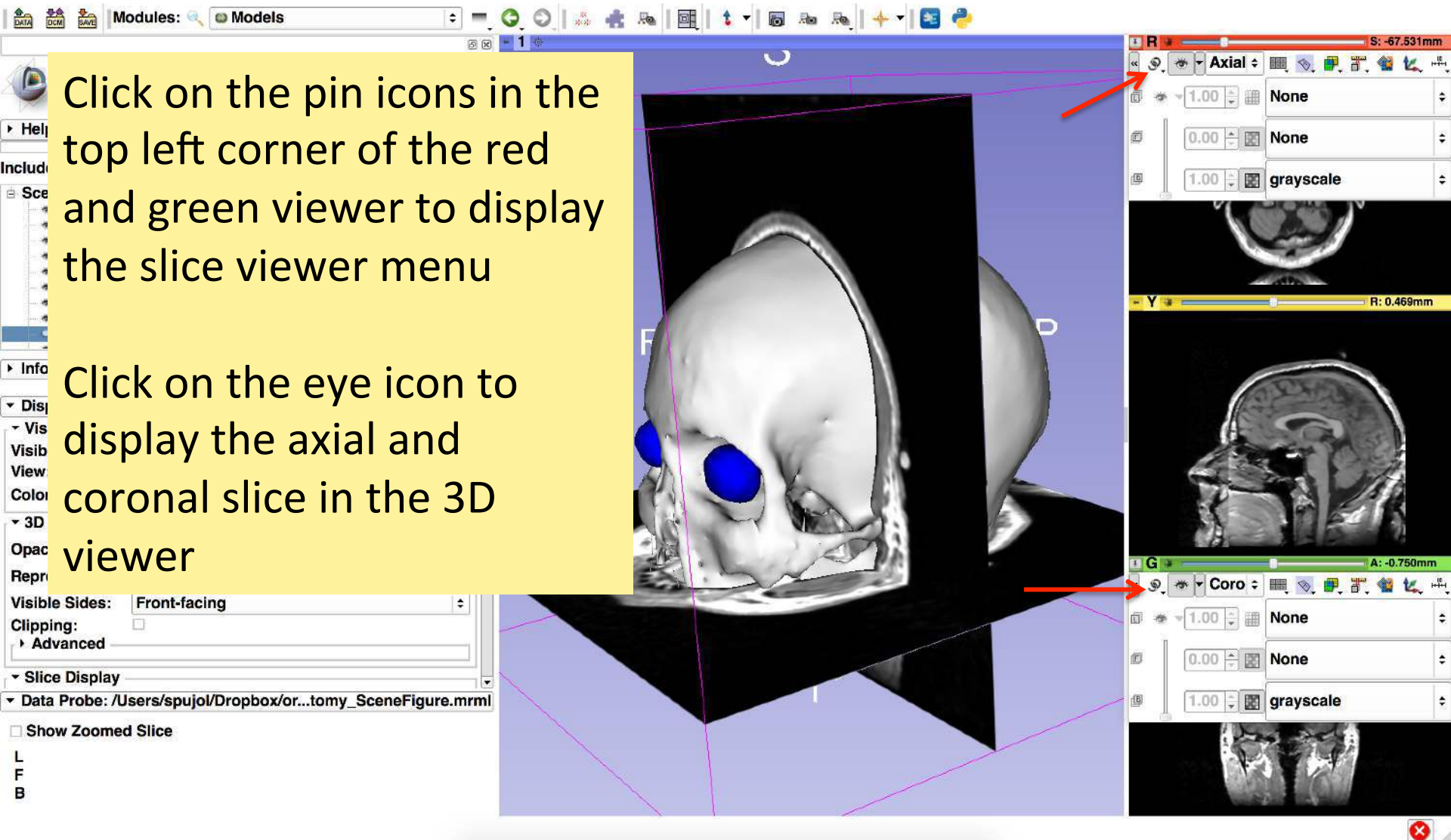
Position the mouse in the 3D viewer, click on the left-mouse button to drag and rotate the model. Click on the right-mouse button to zoom in and out



Anatomical Views

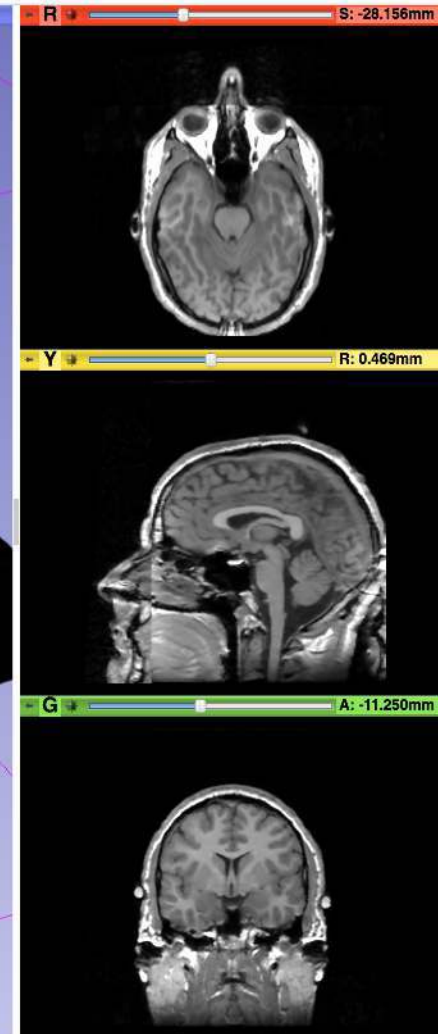
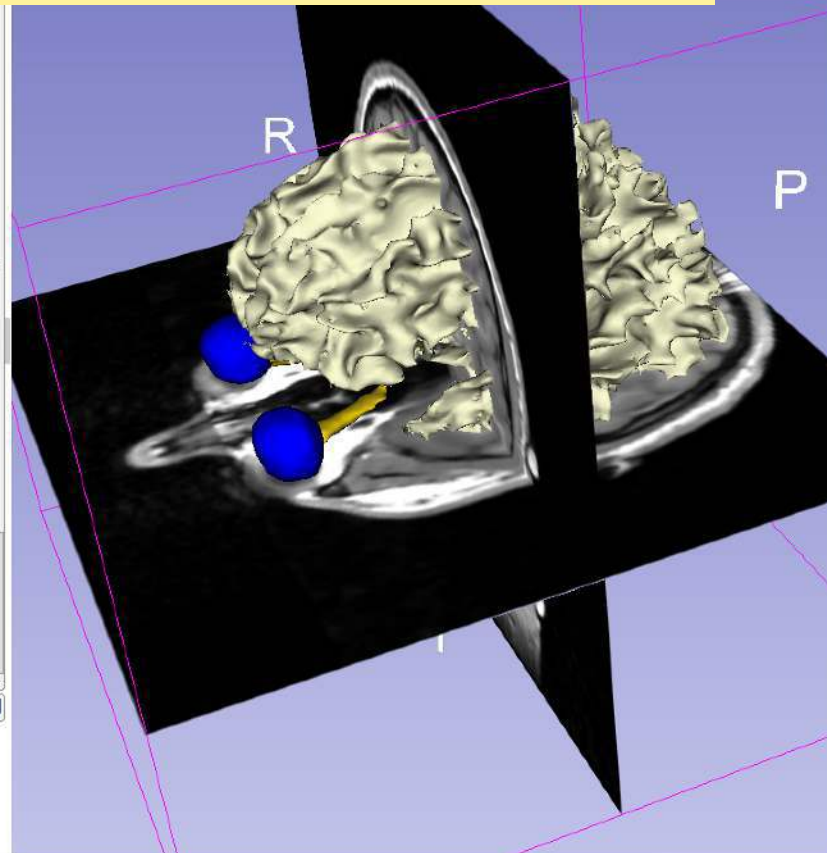
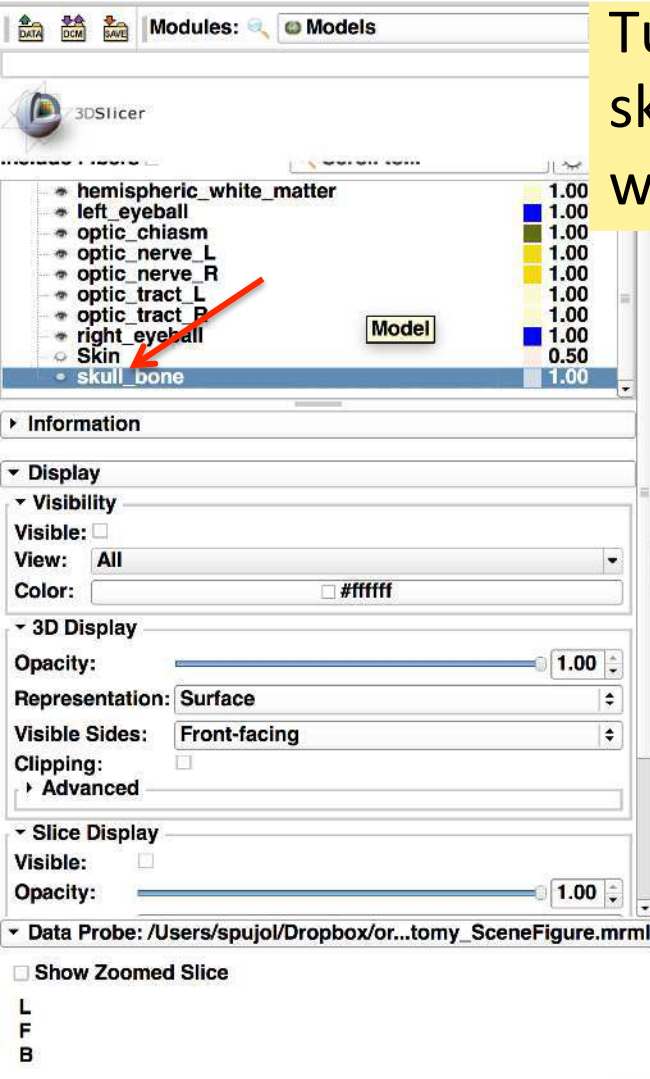
Click on the pin icons in the top left corner of the red and green viewer to display the slice viewer menu

Click on the eye icon to display the axial and coronal slice in the 3D viewer



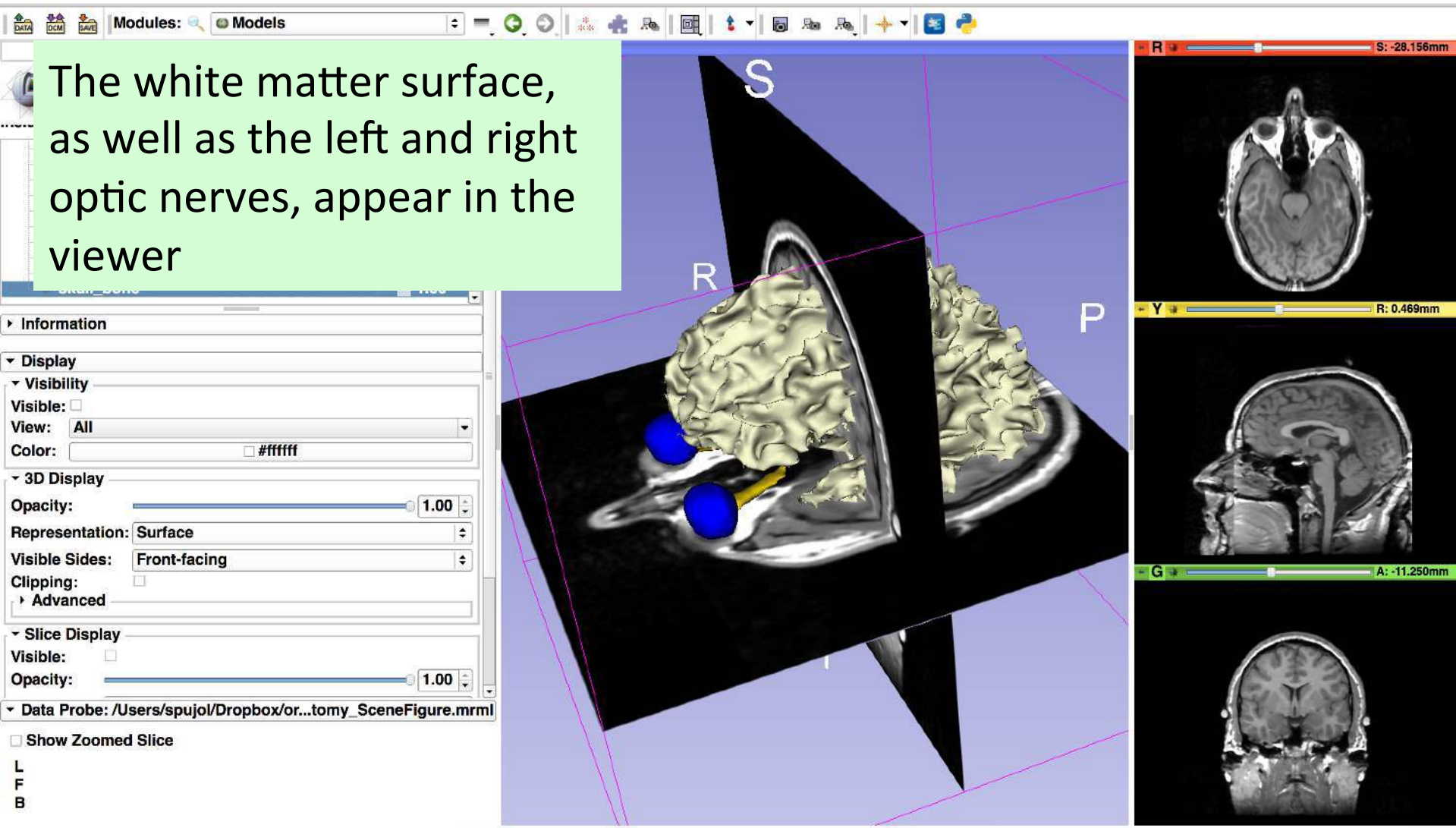
3D Visualization

Turn off the visibility of the skull to display the brain white matter model

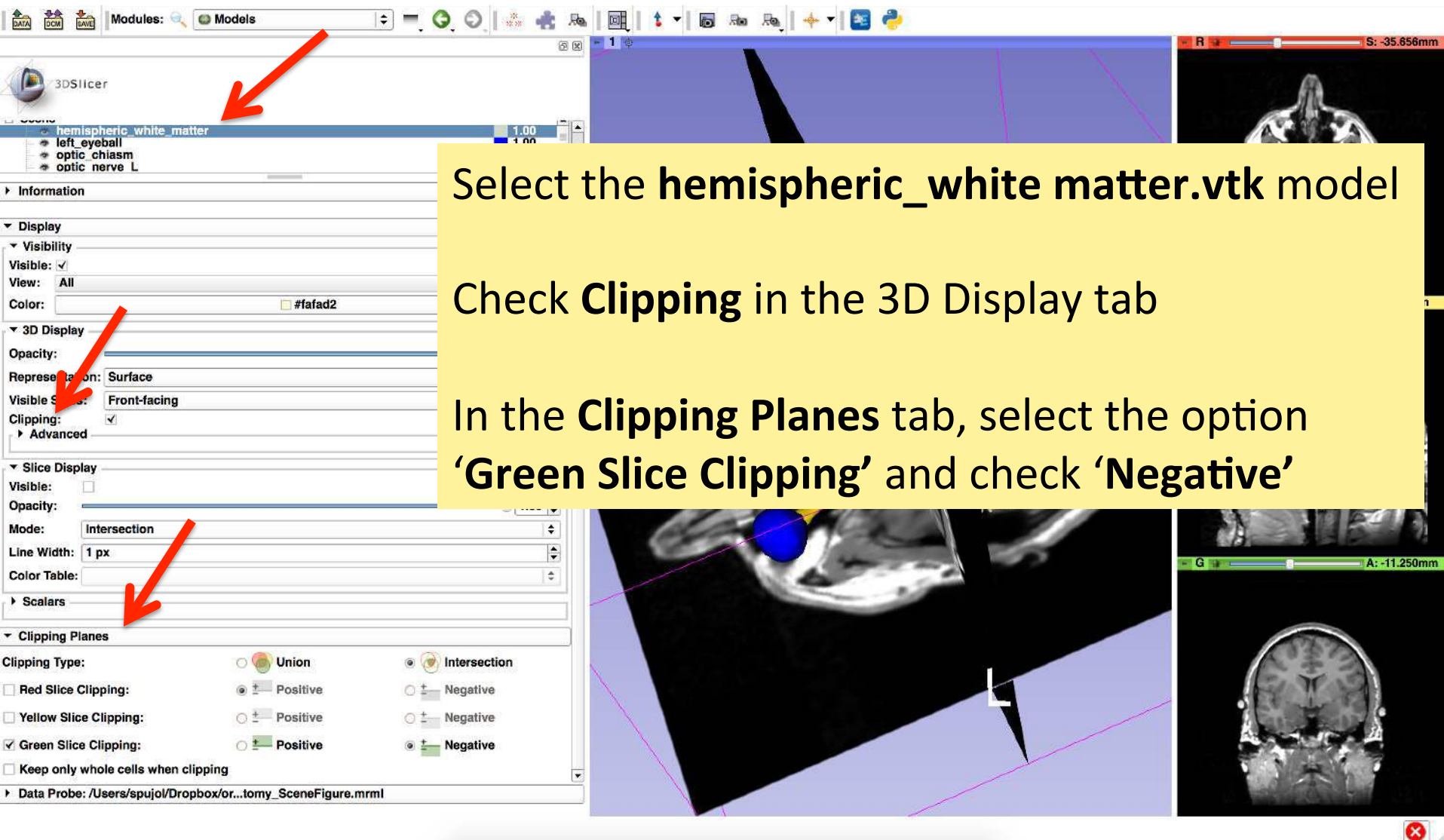


3D Visualization

The white matter surface, as well as the left and right optic nerves, appear in the viewer



3D Visualization



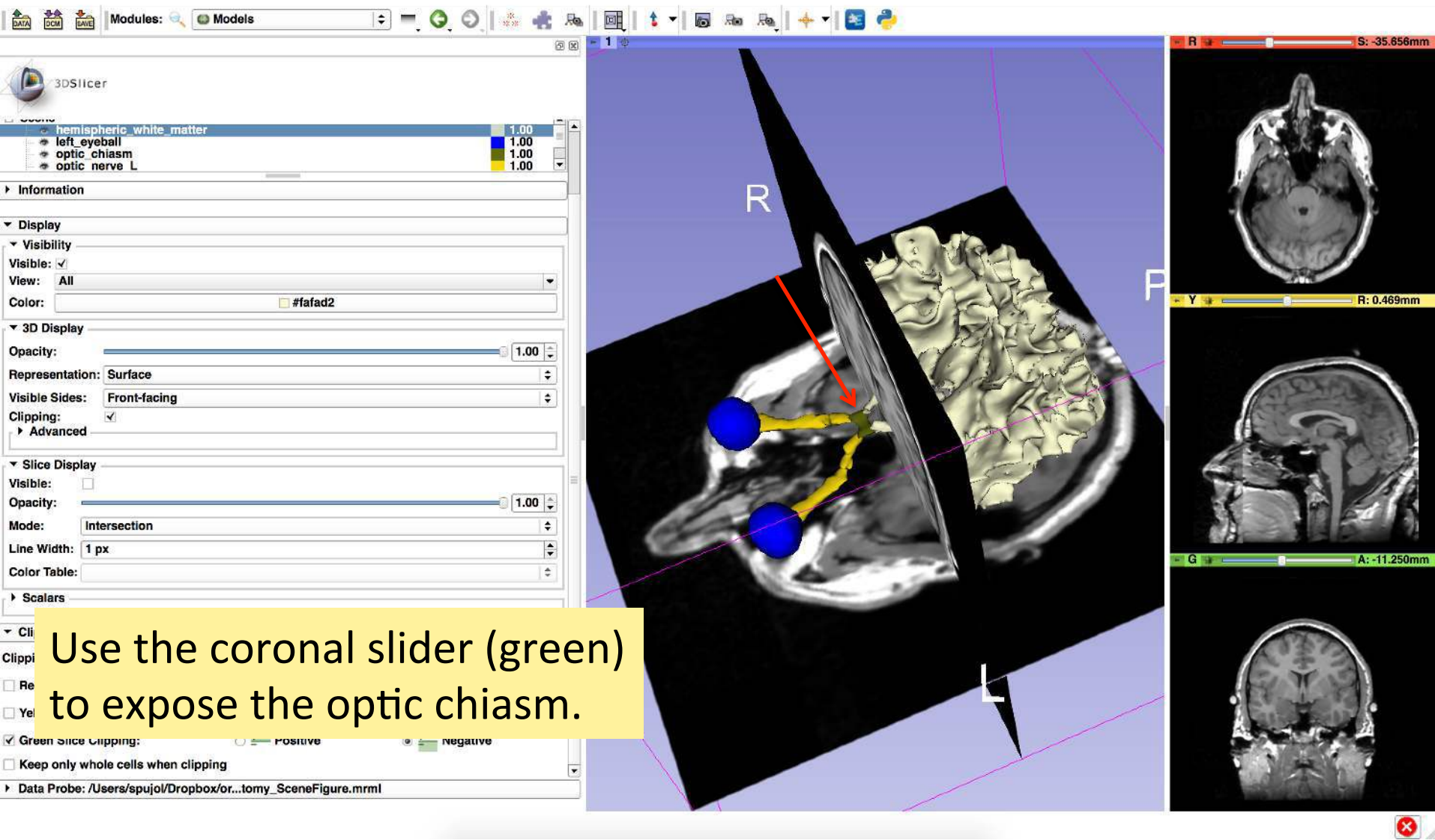
The screenshot shows the 3D Slicer software interface. The 'Models' panel on the left lists several models, with 'hemispheric_white_matter' selected. A red arrow points to this model. The '3D Display' tab is active, showing 'Clipping' checked. Another red arrow points to the 'Clipping' checkbox. The 'Clipping Planes' section at the bottom shows 'Green Slice Clipping' checked and 'Negative' selected. A third red arrow points to the 'Green Slice Clipping' checkbox. The main 3D view shows a brain model with a blue sphere and a black cone. The 'Clipping Planes' section also shows 'Intersection' selected for the clipping type.

Select the **hemispheric_white_matter.vtk** model

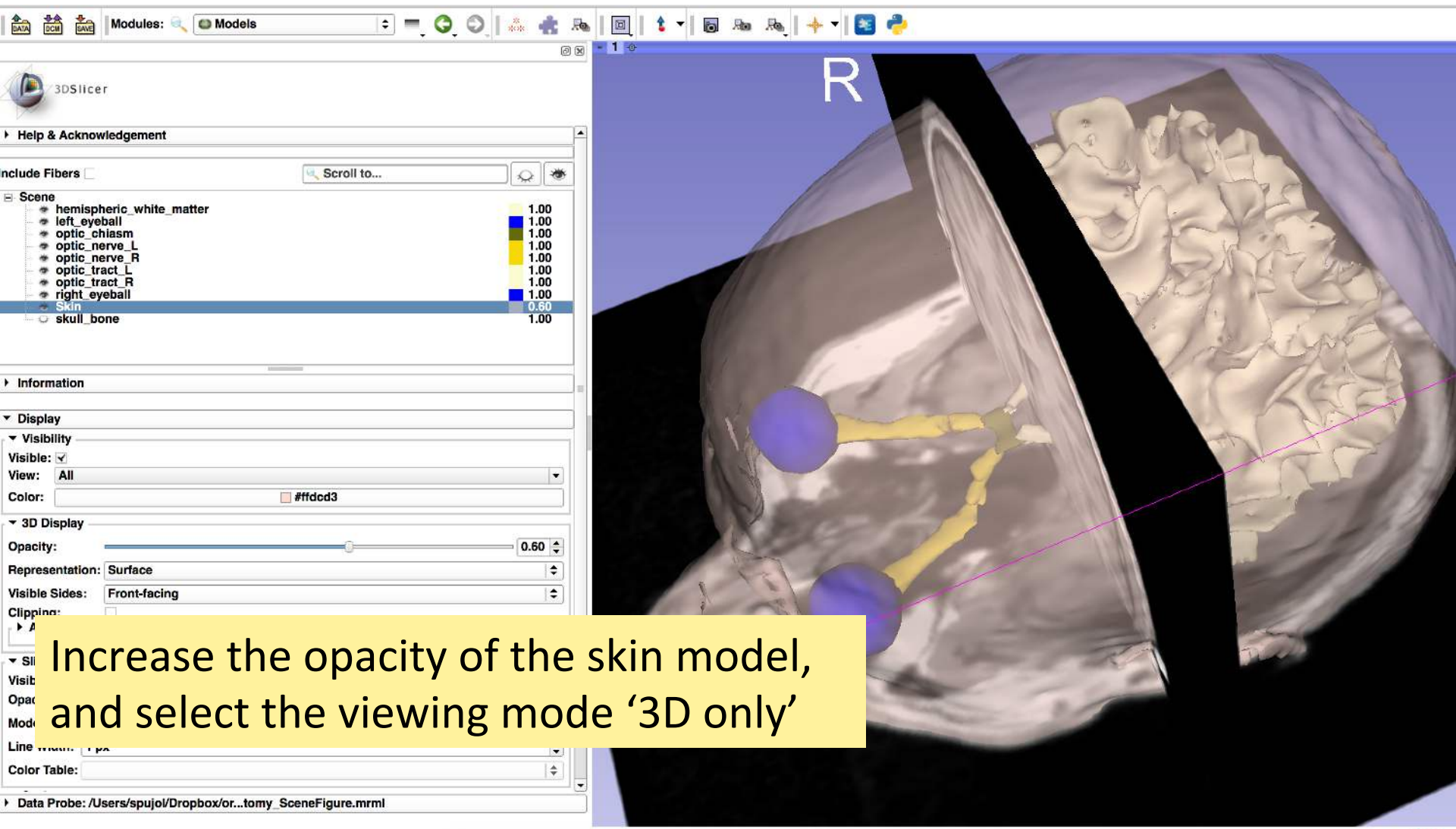
Check **Clipping** in the 3D Display tab

In the **Clipping Planes** tab, select the option '**Green Slice Clipping**' and check '**Negative**'

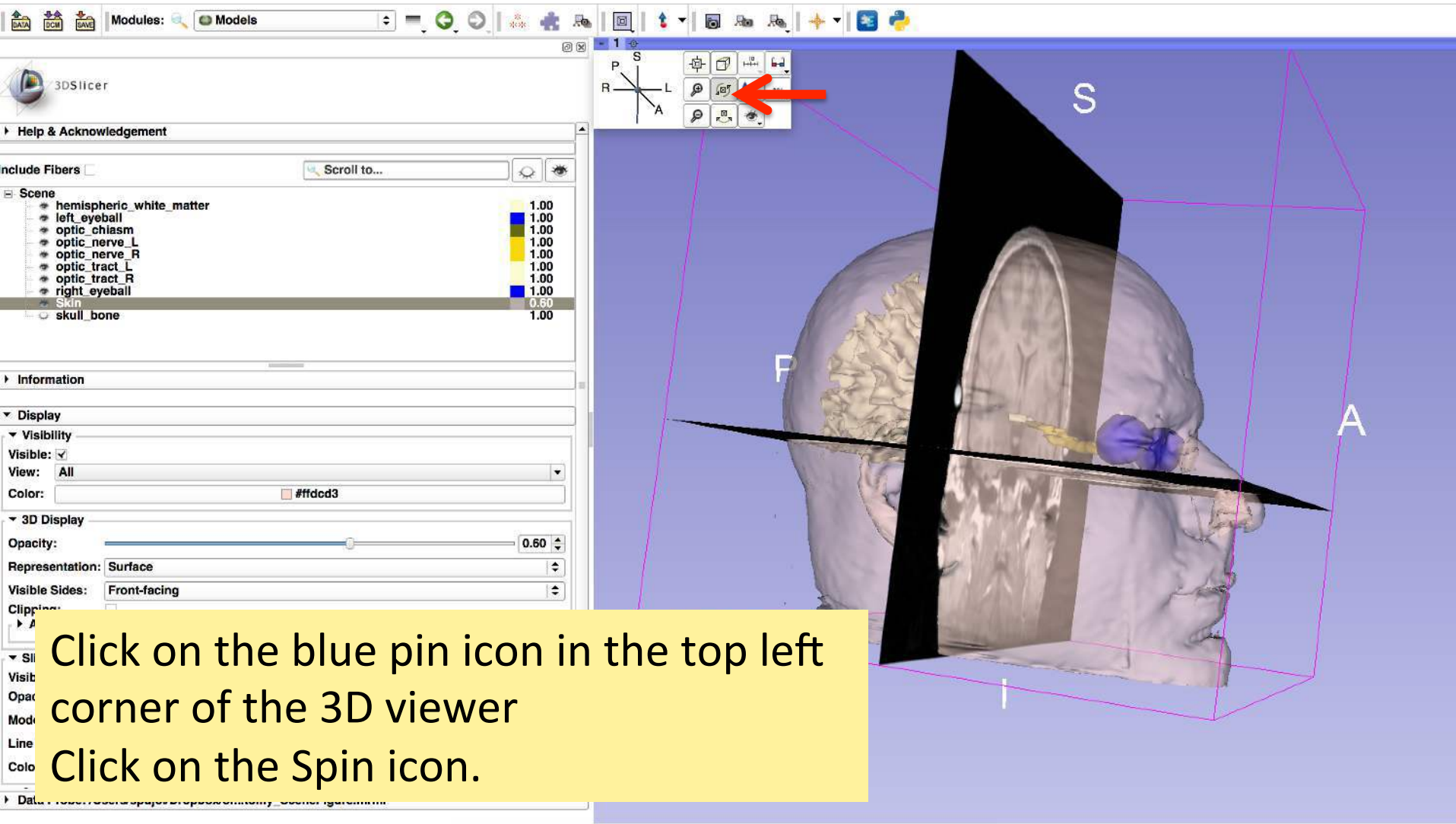
3D Visualization



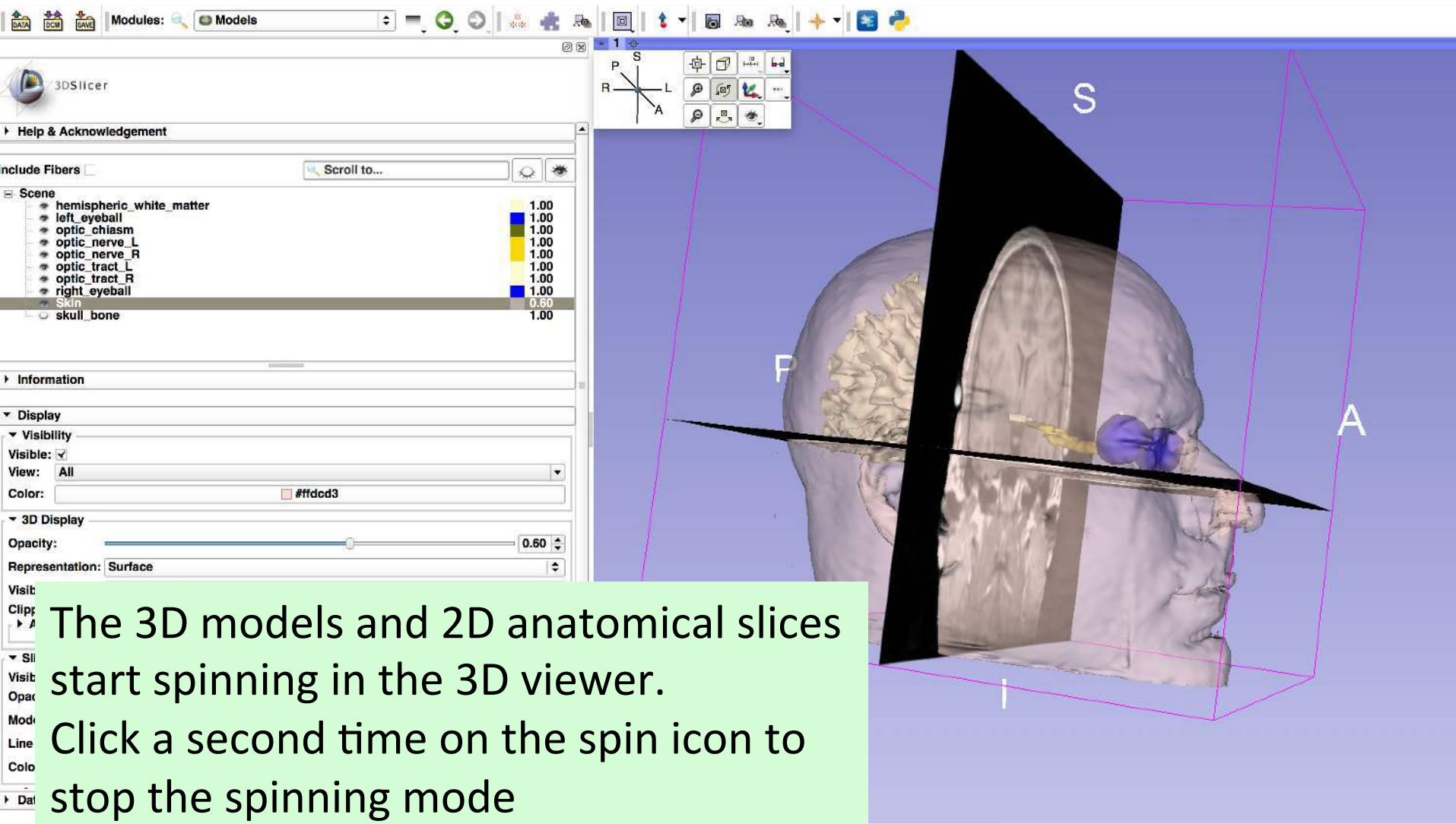
3D Visualization



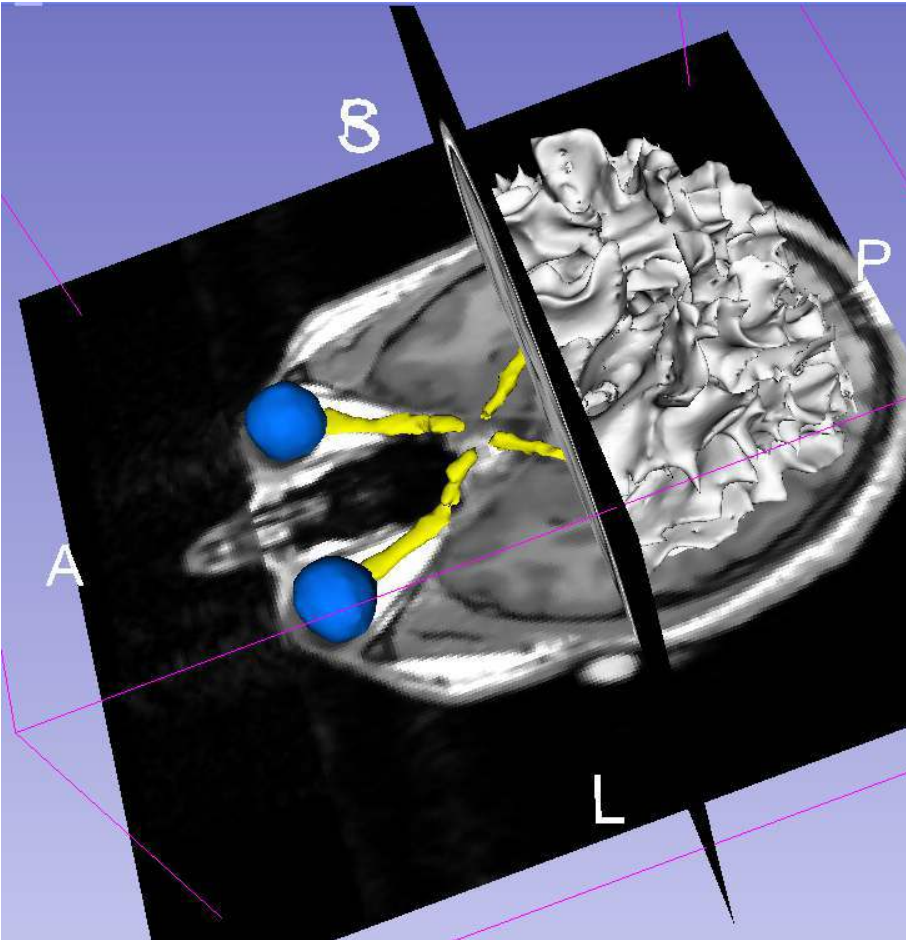
3D Visualization



3D Visualization



Slicer4 minute tutorial



This tutorial was a short introduction on interactive 3D visualization of MRI data and 3D models in Slicer.

The Slicer4 training compendium contains a series of tutorials and pre-computed datasets to learn how to use the software.

Acknowledgments



National Alliance for Medical Image Computing

NIH U54EB005149



Neuroimage Analysis Center

NIH P41EB015902